

The problem was the product

How the modern data stack got lost.

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There is, it seems, a [rough consensus forming](#) about the Apple Vision Pro: The first version won't change your life, but future models might. Nearly every reviewer says that the headset that Apple launched a few weeks ago has too many rough edges and technical inconveniences to revolutionize everyday living—the field of view is too narrow; small text is hard to read; typing is frustrating; it's an intensely solitary experience. However, despite its current limitations, the device [teases the much bolder thing that Apple really wants to build](#): True augmented reality, “where light passes directly through unobtrusive glasses to your eyes, with digital information layered over the top of what you're seeing.” They aren't trying to build an [entertainment console](#); they're building the next “[accompaniment device](#),” meant to be as ubiquitous and as revolutionary as the iPhone. ¹

Notably, Apple hasn't really said any of this. Tim Cook hasn't trotted out a quarterly release schedule; there is no corporate blog post talking about some [master plan](#) of putting [a computer on every face in every home](#). Outside of a [couple](#) sizzle [reels](#), we've all been left to imagine what Vision Pro might become on our own. If the similarity of the reviews are any indication, we're all now anticipating the same future—not by being told what it will be, but by seeing it, on two 4K TVs an inch from our eyes. ²

This, I think, is what makes a truly visionary product. Visionary products aren't just things that seem to reach ahead in time and pull something from the future into the present; they're products that make that future obvious and inevitable to people who couldn't see it before. Visionary products aren't the manifestation of someone's dream; they're products that manifest the same dream in others. They don't just teleport us to a new destination; they hang new stars in the sky, navigating all of us, en masse and in sync, in the same direction.

Put differently, visionary products are [Schelling points](#). ³ Schelling points are meeting points that people congregate towards without any explicit instruction to do so. If you had to meet someone in Paris on Sunday and couldn't make a plan about when and where to do it, the Eiffel Tower at noon is a Schelling point. If two people had to say the same word

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vision Pro is a screening point for an ecosystem of technologists trying to imagine what sort of computing paradigm might be next, and how we might interact with it.

Apple imagined a future; they showed us that future through an incomplete but revelatory product experience; we now understand that future, turning what was a vague and fuzzy and scattered vision of augmented reality into a singular destination. And now, I suspect, thousands of developers are marching us towards that beacon.

Of course, the Vision Pro could still be a flop. That destination could be too ambitious, or not as compelling as people think it might be. And Schelling points don't always work—some at the base, and some at Notre-Dame. ⁴

But it's hard to imagine the Vision Pro being revolutionary *without* engineers and partners and app builders will all chase different i each other's shoulders, [working in rough harmony towards a con](#) scattershot hobbyists and opportunistic founders, building in no

This is one way to think about the convulsive collapse of the cryp the crypto community never produced a coherent vision of what like. There were a lot of vague promises about borderless financi [internet](#). But none of this felt tangible. The problems that the blo theoretical, and hard to locate in our daily lives. And so, no exper companies worked for themselves, building *their* experience rath direction, the ecosystem began to work against itself, fought itse shattered and became a dying echo of what we were all promised

Anyway, speaking of—the modern data stack.

Last week, Tristan Handy [wrote an obituary for the term](#), saying that it's outlived its usefulness: His argument, roughly paraphrased, is this:

1. **The birth:** Almost ten years ago, a handful of companies started building SQL-forward, cloud-native reporting and analytics tools. These products were often cheaper, more accessible, and easier to use than their legacy predecessors.
2. **The brand:** The “modern data stack” became a convenient shorthand for describing these tools, and contrasting them with other products that were from a different era, weren't cloud-native, or focused on other data-based use cases like machine learning and application observability. ⁵
3. **The buzzword:** As the modern data stack grew—partly because it was [genuinely useful](#) and partly because [those who were bankrolling it](#) lost themselves to their [animal spirits](#)—the term started to lose its meaning. Legacy products began to re-architect themselves around the modern data stack's core tenets; investors began describing any investment in data as an investment in the modern data stack; enterprise CIOs began shopping for modern data stack vendors without any awareness of its underlying origins. The modern data stack no longer meant anything specific; “it was just...a meme. A market trend.”
4. **The blow out:** A bunch of new startups, funded by [historically cavalier venture capitalists](#), began chasing the same trend. The data ecosystem filled out, with dozens of companies partnering together to build a collective, complete solution for data and analytics teams.
5. **The blow up:** In 2022, the market turned. Funding rounds evaporated with rising interest rates, customers became much more cost-conscious, and Silicon Valley began hyperventilating [about AI](#) ⁶ and [not the modern data stack](#).



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6. **The burial:** And so, he concludes, we should bury the term, and rebrand this whole situation as the analytics stack. “The cloud has won; all data companies are now cloud data companies. *Let’s move on.*”

Tristan is a [more careful historian](#) and [more central character](#) in the modern data stack’s story than I am, so I can’t disagree with his narrative too much. But as a [bigger cynic](#) than him, I think he omits one very key element of this story: The problem wasn’t the market around the products we were building; *the problem was the product itself*.

In the modern data stack’s early days, most vendors were trying to replicate what already existed, but in ways that were bigger, faster, and cheaper. There was a fairly clear experiential Schelling point then—make data practitioners lives better by recreating Oracle, MicroStrategy, Informatica, SAS, and a handful of other geriatric, proprietary, and expensive data tools using the cloud, open-source languages, and consumer-grade user interfaces. Fivetran was Informatica and SnapLogic, but [automated and simple](#). “Snowflake [saw an opportunity](#) to modernize [the data warehouse] by reducing the complexity and putting the whole thing in the cloud.” Mode’s early fundraising decks [compared it to SAS](#), but powered by SQL and Python.

It’s what happened next when things went wrong. Tristan describes the explosive phase that followed—the blow out, in the narrative above—as a largely positive period of [dynamic innovation and fruitful coordination](#):

The end-to-end problem was far too big for any one startup to solve, and so swim lanes were established and partnership ruled the day.

There was a lot of valuable co-marketing, partnership deals, co-sponsored events, and co-selling. This had real value for everyone involved—customers and vendors alike. Companies voluntarily integrated their products together, cross-promoted each other publicly, and built partnerships that made owning and operating these technologies far easier for customers. ...

This was a beautiful thing. Private capital fueling founders, who not only built their own products but were consciously coming together to build an ecosystem, leading to the rapid buildout of interoperable products all taking advantage of a new technology platform (the cloud).

That’s not quite how I would describe what happened? As it grew, the ecosystem didn’t continue marching towards the vision that inspired it in its early years; it completely lost its bearings. The unifying experience that guided the original modern data stack—old tools, made easy—got replaced with a bunch of catchphrases about instant insights, better decisions, and data democratization—i.e., borderless financial markets, smart contracts, and [a decentralized internet](#). These ambitions dramatically expanded the modern data stack’s scope from streamlining how data teams were already working to transforming how entire companies should operate.

But there was no agreement on *how* we’d do this. There was no visible picture of the collective experience the modern data stack was supposed to create for people who weren’t data practitioners. So, everyone tried to figure out how to “democratize data” or “empower decision-makers” on their own—and it turned into a mosh pit. Did documentation live in a BI tool or a data dictionary? In whichever one you were selling. Were we replacing spreadsheets, or integrating with them? Depends on if you thought you could compete with Excel. How would an executive know if they can trust the dashboard they’re looking at? By logging into your product, probably. There was no Schelling point to build towards; there were skills, pointing towards whatever thing they had built.

Moreover, we got distracted by circular problems of our own making. We created pipelines to shuffle data around, and orchestrators to coordinate those pipelines, and observability dashboards to monitor the orchestrators, and incident

Though the economics of the modern data stack were always tenuous,⁷ I don't think that Jerome Powell or Sam Altman are who did us in.⁸ The problem was that the modern data stack's ambitions expanded—from helping data teams to revolutionizing companies—with no communal vision for how to get there. There was money, and ambition, and loosely shared philosophy that favored SQL and the cloud, but there were [no answers to these questions](#):

It's become popular to say that data teams should think of everything they create as a product, and the rest of their colleagues as their customers. To build on this idea, what should that product be? What should it feel like to go from question, through technology and tools, through collaboration and conversation, to an answer?

Our current solution, though full of pieces that are individually great, is a disjointed one. Some responsibilities, like metric governance, are shared across multiple tools; others, like tracking what happens after a question is answered, are [mostly ignored](#). Even seemingly basic questions, like "where do I first go to ask a question?," don't have clear answers. ...

How should it feel to use modern data stack? What can the system add up to? What's the best way for people to answer an unfolding series of questions, to trust those answers, and to decide what to do next? What can we create for people who care about how well the modern data stack works, regardless of where the lines are drawn between products and services?

If you want to go fast, the proverb says, go alone. If you want to go far, go together.

Modern data stack vendors chose speed, and never attempted to truly build something together. Companies built product integrations as a way to cross-sell, or to paper over feature gaps, or to ingratiate themselves with more successful vendors.⁹ But most partnerships were temporary and transactional agreements of mutual convenience that went [co-sponsored afterparty](#) deep.

One way or another, we weren't going to last by going it alone. We were going to have to develop a coherent vision as a bunch of independent companies, or all get acquired and be handed that vision by a few big companies. So far, the latter seems to be winning.

—

Still, a bigger question remains: *Why* weren't we able to produce a more coherent vision for what the modern data stack was supposed to become? Why has nobody been able to quite articulate *how* data products go beyond just streamlining self-referential workflows, and become the transformative strategic asset that they proudly proclaim to be?

One answer is because they can't. [Data may not have that revolutionary power](#). It may simply be dryly utilitarian.

If that's true, Tristan's proposal for rebranding the modern data stack as the analytics stack would be a fitting coda. It both reorients the ecosystem back towards its original goals of serving the needs of practitioners and away from the buzzy neologisms that defined the last few years, and it has a sort of dull, workmanlike tone that will [bore thrill seekers](#) and attract [people who want to solve real problems](#). Whereas the modern data stack was expansive—it could be a boundless platform; it could shape-shift its way into being a critical part of the [\\$7 trillion dollar AI market](#)¹⁰—analytics is a function. It is [accounting](#): Small and localized and bound. And [frugal](#):

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needing to build a CRM for a nascent sales team, and having to buy a tool for logging phone calls, a tool for logging emails, a tool that stores these logs, a tool that helps sales reps see the status of their opportunities, a tool that helps sales leaders keep track of their team's performance, and a tool to alert you when other tools break. While products like these may get hung off of Salesforce at some point, they're bought by growing sales teams, not integral legs to an eight-legged stool.

The other answer, however, is we simply built the modern data stack in the wrong way. It's possible that the transformative potential of data is real; we just lost sight of how to get there amid the riot. If we can figure out that vision—if someone can reveal their dream to all of us, in a way so visceral we can dream it too—we can start chasing the same thing again, as we did a decade ago, when the dream was smaller.

In other words, the bigger target—not just analytics, but organizational transformation—remains. The space got invaded by hordes of new companies, and economic shock is nuking a lot of them out of existence. [But the target remains.](#)

Lost in the sauce or lost in the game?

Another—also cynical—way to think about the modern data stack is through the eyes of the people who created it. In a recent post, Anu Atluru proposes that there are [two types of entrepreneurs](#). Those [lost in the sauce and those lost in the game](#):

There are those in it for the love of the mission that they're obsessed with championing. And there are those in it for the love of the game of entrepreneurship itself, almost irrespective of the mission of the business. *Sometimes the mission is the business and sometimes the business is the mission. Type I, and Type II.*

The data world, I'd argue, has very few true mission-driven, Type I founders. At best, most data founders are motivated by bringing some elegance to an ugly technical problem; at worst, we were opportunists who convinced ourselves we could turn some [internal tool](#) into a business, and saw founding a company as a way to avoid the career cul-de-sac of [becoming a mid-level data director](#). There aren't many people who start data companies simply to manifest something in the world. For most of us, it's important that we manifest it, either because we'll have fun building it or because we want the rewards for doing so.

As Anu says, there's nothing wrong with starting a company for those reasons; companies can exist for noble reasons, and they can exist to [make something great that makes a lot of money](#). Both are fine.

However, type II founders aren't great when an ecosystem needs to *collectively* build something great, which is what the modern data stack was. Had we all been chasing a mission, we might've figured out ways to help everyone get there. But when we're mostly there for the game, we care more about the individual role we're playing than the communal thing we're building.

- 1 Although, I think there's a very real possibility that the Vision Pro becomes an [engulfing](#) device. Because people are definitely going to get addicted to this thing, right? According to one review, wearing a Vision Pro is [like dreaming](#). You can conjure entire worlds—soon, [from nothing](#), literally, with [obvious implications](#)—and drown yourself in them. And, sure, [they may be nothing more](#) than “hollow imaginings meant to augment reality through a laminated veil of glass and sensors.” But what if your actual reality is already empty and isolated? What if detaching isn't a downside, but exactly the point?

If millions of people already sink themselves into our crude [Experience Machines](#)—drugs, porn, video games—and disappear, what hope do we have against the real thing?

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- 4 Complete tangent, but—[The Bells of Notre Dame](#) and [Hellfire](#) are two underrated (and, like, [definitely not G-rated](#)) Disney bangers.
- 5 For example, I don't think anyone's ever considered [Scale AI](#) or [Honeycomb](#) as part of the modern data stack.
- 6 There's a lot to love about that TikTok, but the part at 0:53 where her sister sticks the card in the frame and says, "what does it say?"—absolute gold.
- 7 And predictably so. From this blog, [in the fall of 2021](#):

The ecosystem is creating category after category, throwing out product after product. Thus far, the market has absorbed them—but the dynamic is unstable. With few dominant players and fewer agreed-upon categorical standards, customers are choosing from a large inventory of small companies, offering dozens of startups enough of a foothold to reasonably claim that they provide more value than they cost. Furthermore, because of the variance in buyers and business models, the true cost of the stack is amortized across different teams and obfuscated by irregular billing logistics.

But data teams and IT departments have budgets. Even if every product is worth it on its own, the collective cost eventually becomes too much to bear. Something has to give.

- 8 Here's another way to ask this question. Had the economy stayed hot and ChatGPT stayed buried inside OpenAI, would the modern data stack, as it existed in 2021, have survived? My vote: No.
- 9 What the LinkedIn post says: "We're proud to announce that we're a new Amazon Select partner! We're excited to work with the Redshift team to continue building great experiences for our customers! #data #Redshift #AWS #CantStopWontStop"

What the LinkedIn post means: "The only coordination between us and Redshift will be periodic emails from their partner solutions manager reminding us that we need to upgrade our Redshift drivers if we want to keep our listing in AWS Marketplace. And we only did this because we've missed our pipeline targets for the last three quarters. We know that this won't fix it, but we can show our increasingly anxious board that we're doing something, and that might make us feel good for a minute."

- 10 I mean, this is obviously stupid, but just so we're clear: A \$7 trillion investment vehicle isn't a VC fund; it's a [centrally planned economy](#). And that's not suddenly a good thing, [just because Sam Altman is your guy](#).

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Ari Bajo

Feb 17, 2024 Edited

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MDS tools have been solving the same technical problems on their own. Notably, integrations. Data movement, orchestration, transformation, quality, bi tools... all need integrations for each data storage system... And each tool solved it from scratch, with no common ground. Let me explain, when I worked for a BI tool, we wrote integrations to read data from data storages. When I worked for a data movement tool, we wrote connectors to read and write data from data storages. When I worked for a data quality tool, we wrote connectors to execute queries (tests) on data storages. Didn't work for a data orchestration tool, but operators basically make API calls to external systems, notably data storages. The value is in the data, or metadata if the system is not a data storage per se. And the common thing, is making api calls to these systems: read data, write data, query data (execute tests), trigger actions (orchestration). If there was a common framework to interact with external systems, then I think data tool companies could be more ambitious with a clear vision focused around data. We are missing standards, and a vision like you said. A standard to connect to any external system and be able to read, write, query and trigger actions. Then we need user experiences on top of that. And probably also a common standard on what to build user experiences.

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3 replies by Benn Stancil and others



Nicole Edmonds

Feb 19, 2024 Edited

♥ Liked by Benn Stancil

Also, any outsider can look at Matt Turck's Machine learning AI, and data landscape and think - how in the, what?, am I to make sense of this? There's no connections illustrated between all of these products, and even if there was, it would look like every conspiracy theorist's pictures with string mood board. Hard to make sense of any gathering point in the chaos indeed.

My search turned up 2021's - how many products can one person know, have skills in, and even further, know now to fit it all together? <https://mattturck.com/wp-content/uploads/2021/12/2021-MAD-Landscape-v3.pdf>

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